Classification Metrics

Accuracy

Accuracy measures the proportion of correctly classified instances (both true positives and true negatives) out of the total instances. It is computed as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}. \tag{4}$$

Sensitivity

Sensitivity quantifies the models' ability to correctly identify positive instances. It is calculated as:

$$Sensitivity = \frac{TP}{TP + FN}. \tag{5}$$

Specificity

Specificity measures the models' ability to correctly identify negative instances. It is computed in the following way.

$$Specificity = \frac{TN}{TN + FP}. \tag{6}$$

Precision

Precision assesses the proportion of true positive predictions among all positive predictions. It is given by:

$$Precision = \frac{TP}{TP + FP}. \tag{7}$$

NPV

NPV quantifies the proportion of true negative predictions among all negative predictions. It is calculated as:

$$NPV = \frac{TN}{TN + FN}. \tag{8}$$

F1 Score

The F1 score is the harmonic mean of precision and sensitivity, providing a balanced measure of the models' performance. It is computed as:

$$F1\ score\ =\ 2 \times \frac{Precision \times\ Sensitivity}{Precision + Sensitivity}. \tag{9}$$

DMQ Impairment Scale Items

These items come directly from the Duke Misophonia Questionnaire Impairment Subscale (Rosenthal et al., 2021).

For the following section, please use the scale below:

0	1	2	3	4
Not at all	A little	Moderately	Quite a bit	Extremely

Please rate the extent to which the bothersome sound/sounds and your reactions to them negatively affected the following in the past month, on average.

1. My ability to be with other people.
2. My performance at work or school.
3. The quality of my romantic relationships.
4. My ability to function in daily activities without help.
5. How much I enjoy spending time with my family.
6. My ability to work with others.
7. My self-esteem.
8. My ability to maintain employment.
9. The quality of my relationships with my friends.
10. How connected I feel to other people.
11. My ability to live with other people (e.g. roommate, partner).
12. My ability to "be myself".

Supplemental Figures

Supplemental Table 1. *Marginal Means for all approaches, including the RF approach using predicted probabilities and ROC cut-point selection instead of a Bayes Classifier across groups and type of condition.*

	Marginal Means (SD)					
	Non-DIF Conditions					
	Accuracy	Sensitivity	Specificity	Precision	NPV	F1 Score
IRT						
Full Sample	.96 (.01)	.96 (.02)	.96 (.03)	.99 (.01)	.96 (.02)	.97 (.01)
RF Class						
Full Sample	.34 (.09)	.42 (.09)	.01 (.10)	.61 (.17)	.42 (.09)	.50 (.10)
RF Prob						
Full Sample	.34 (.09)	.42 (.09)	.01 (.10)	.61 (.17)	.42 (.09)	.50 (.10)
	Non-Visible DIF Conditions					
	Accuracy	Sensitivity	Specificity	Precision	NPV	F1 Score
IRT						
Full Sample	.94 (.05)	.93 (.07)	.94 (.06)	.98 (.02)	.93 (.07)	.96 (.04)
Reference Group	.92 (.09)	.91 (.11)	.97 (.04)	.99 (.01)	.91 (.11)	.95 (.07)
Focal Group	.95 (.06)	.97 (.03)	.84 (.26)	.96 (.07)	.97 (.03)	.96 (.04)
RF Class						
Full Sample	.96 (.02)	.97 (.02)	.89 (.06)	.97 (.01)	.97 (.02)	.97 (.01)
Reference Group	.95 (.02)	.96 (.03)	.91 (.07)	.98 (.01)	.96 (.03)	.97 (.02)
Focal Group	.96 (.03)	.98 (.02)	.82 (.18)	.96 (.04)	.98 (.02)	.97 (.02)
RF Prob						
Full Sample	.34 (.09)	.40 (.12)	.04 (.18)	.60 (.20)	.40 (.12)	.50 (.10)
Reference Group	.36 (.10)	.43 (.13)	.04 (.18)	.62 (.20)	.43 (.13)	.52 (.11)
Focal Group	.30 (.12)	.37 (.16)	.04 (.19)	.58 (.21)	.37 (.16)	.45 (.14)
	DIF with Visible DTF Conditions					
	Accuracy	Sensitivity	Specificity	Precision	NPV	F1 Score
IRT						
Full Sample	.91 (.08)	.91 (.10)	.91 (.10)	.97 (.04)	.91 (.10)	.93 (.06)
Reference Group	.88 (.14)	.86 (.18)	.98 (.04)	.99 (.01)	.86 (.18)	.91 (.14)
Focal Group	.93 (.09)	.98 (.03)	.67 (.38)	.93 (.10)	.98 (.03)	.95 (.06)
RF Class						
Full Sample	.95 (.02)	.96 (.02)	.89 (.06)	.97 (.01)	.96 (.02)	.97 (.01)
Reference Group	.95 (.02)	.95 (.03)	.92 (.07)	.98 (.01)	.95 (.03)	.97 (.02)
Focal Group	.95 (.03)	.98 (.02)	.76 (.23)	.95 (.05)	.98 (.02)	.97 (.03)
RF Prob						
Full Sample	.33 (.09)	.39 (.14)	.07 (.24)	.58 (.23)	.39 (.14)	.50 (.11)
Reference Group	.39 (.11)	.44 (.17)	.07 (.23)	.63 (.24)	.44 (.17)	.55 (.12)
Focal Group	.23 (.12)	.31 (.19)	.07 (.25)	.52 (.24)	.31 (.19)	.37 (.15)

Note: DIF = differential item functioning.

Supplemental Table 2. *Effect Sizes and Marginal Means: Main effects and interactions for classification outcomes in conditions with DIF that is not visible in the TIF.*

	Partial η^2					
	Accuracy	Sensitivity	Specificity	Precision	NPV	F1 Score
Method	0.08	0.21	0.35	0.21	0.21	0.08
Sample Size (N)	–	–	0.07	–	–	–
Number of Items	0.09	–	–	–	–	0.08
Prevalence (p)	0.10	0.09	0.10	0.30	0.09	0.24
DIF Rate (d)	0.12	0.07	–	–	0.07	0.11
b Strength (b)	0.12	0.07	–	–	0.07	0.12
p × Method	–	–	0.18	–	–	–
$d \times b$	0.15	0.09	–	–	0.09	0.14
d × Method	0.07	–	–	–	–	0.06
b × Method	0.09	0.07	–	–	0.07	0.09
d × Method × b	0.12	0.08	–	–	0.08	0.11
	Marginal Means (SD)					
	Accuracy	Sensitivity	Specificity	Precision	NPV	F1 Score
IRT						
Full Sample	.94 (.04)	.94 (.05)	.95 (.05)	.99 (.02)	.94 (.05)	.96 (.03)
Reference Group	.94 (.06)	.93 (.08)	.97 (.03)	.99 (.01)	.93 (.08)	.96 (.06)
Focal Group	.95 (.04)	.97 (.03)	.91 (.14)	.98 (.04)	.97 (.03)	.97 (.03)
RF						
Full Sample	.96 (.02)	.97 (.02)	.89 (.06)	.97 (.01)	.97 (.02)	.97 (.01)
Reference Group	.96 (.02)	.96 (.02)	.91 (.07)	.98 (.01)	.96 (.02)	.97 (.01)
Focal Group	.96 (.02)	.98 (.02)	.86 (.12)	.97 (.03)	.98 (.02)	.97 (.02)

Note: Partial η^2 = effect size on the give outcome; DIF = differential item functioning; TIF = test information function; FR Ratio = the ratio of the focal group sample size to the reference group sample size. IRT = item response theory-based classification; RF = random forest classification using a bayes classifier. Effect sizes < 0.06 (medium effect; Cohen, 2009) were omitted. In such cases when a main effect or interaction was not found to be influential (effect sizes < 0.06) on any classification measure, it was omitted from this table for simplicity. All results from the random forest approaching using probabilities were not included in the ANOVA sample, as they were shown to perform notably worse than other approaches upon visual inspection.

Supplemental Table 3. *Demographic Information for the Empirical Example*

Variable	N (%)
Gender	
Male	143 (19.7%)
Female	582 (80.3%)
Education	
Less than high school	2 (0.3%)
High school	376 (51.9%)
Some college	339 (46.8%)
Vocational training	2 (0.3%)
Associate degree	3 (0.4%)
Bachelor's degree	2 (0.3%)
Master's degree	0 (0%)
Professional degree	0 (0%)
Doctorate degree	0 (0%)
Prefer not to answer	1 (0.1%)
Ethnicity	
American Indian or Alaskan Native	8 (1.1%)
Asian	58 (8.0%)
Black	42 (5.8%)
LatinX	62 (8.6%)
Native Hawaiian or Other Pacific Islander	1 (0.1%)
Other	6 (0.8%)
White	548 (75.6%)
Misophonia (defined by MQ severity score)	
No	594 (81.9%)
Yes	131 (18.1%)

Note: MQ = Misophonia Questionnaire

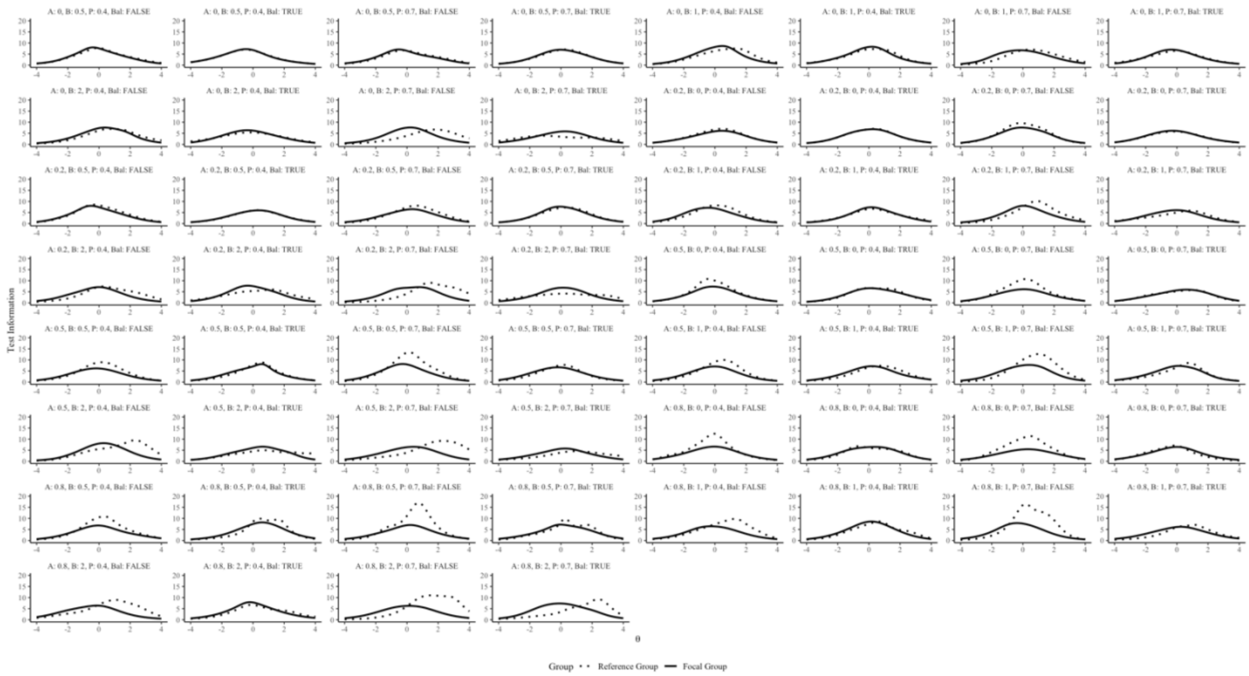
Supplemental Table 4. Variable Importance for Predicting High Impairment Using Random Forest Classification

Item	Importance	% Total
1	30.18	27.54
11	18.88	17.23
6	18.74	17.1
5	13.72	12.52
12	11.92	10.88
9	7.96	7.26
8	6.27	5.72
3	4.4	4.01
7	3.64	3.32
10	3.14	2.87
2	0	0
4	0	0

Note: This table displays the variable importance scores generated from a weighted random forest model trained to classify participants into diagnostic groups for misophonia (1 = have misophonia, 0 = does not have misophonia), using item-level responses from the impairment subscale of the Duke Misophonia Questionnaire (Rosenthal et al., 2021). Importance scores represent the decrease in model accuracy when each item is permuted, and the "% Total" column indicates each item's relative contribution to the total variable importance across all predictors. Bolded items indicate those that exhibited gender-based differential item functioning (DIF), identified via prior IRT-based DIF analysis and marked here for interpretive clarity (Items 1, 2, 8, and 11). Items with negative importance values (e.g., Item 4) may reflect noise or suppression effects in the model. As such, we recoded their importance value to be zero to indicate that they were not valuable to the model. Higher values denote greater predictive contribution to the classification outcome. Importance scores were rounded to two decimal places and sorted in descending order.

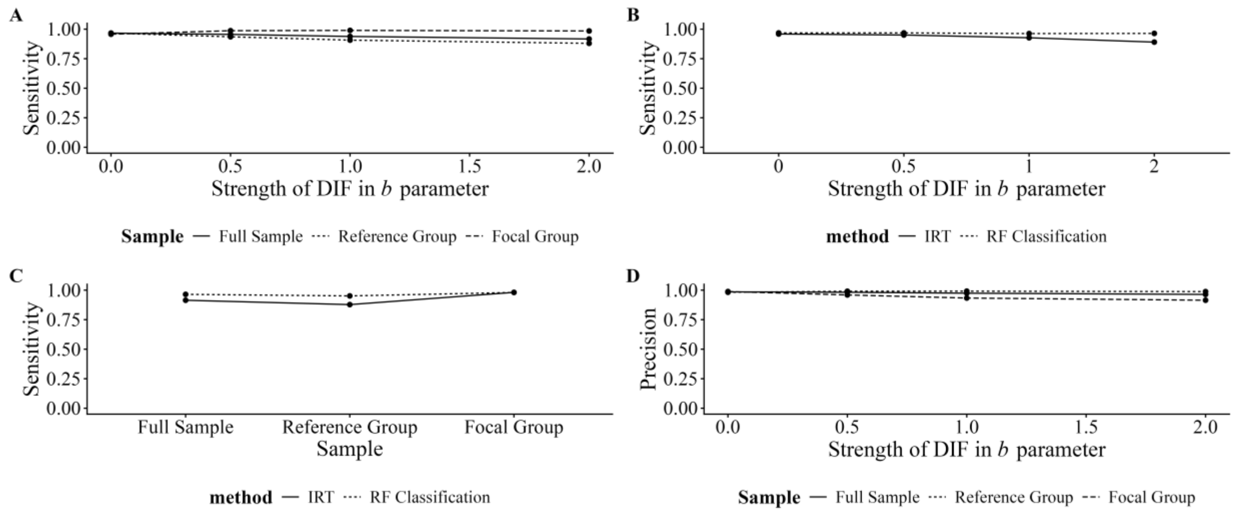
Supplemental Figure 1.

Comparison of Test Information Functions (TIFs) between reference and focal groups across different Differential Item Functioning (DIF) conditions.



Note: The facets display results for varying DIF effect types (discrimination [A], difficulty [B], and both), DIF effect magnitudes (A: 0.2, 0.5, 0.8; B: 0.5, 1, 2), DIF proportions (P: 0.4, 0.7), and DIF balance conditions (balanced and unbalanced). Solid lines represent the TIF for the reference group, while dotted lines represent the TIF for the focal group. The horizontal axis represents theta (θ), or the latent ability scale, and the vertical axis represents the level of test information. Each panel corresponds to a unique combination of DIF parameters, demonstrating how different DIF conditions affect the TIF of the focal group compared to the reference group.

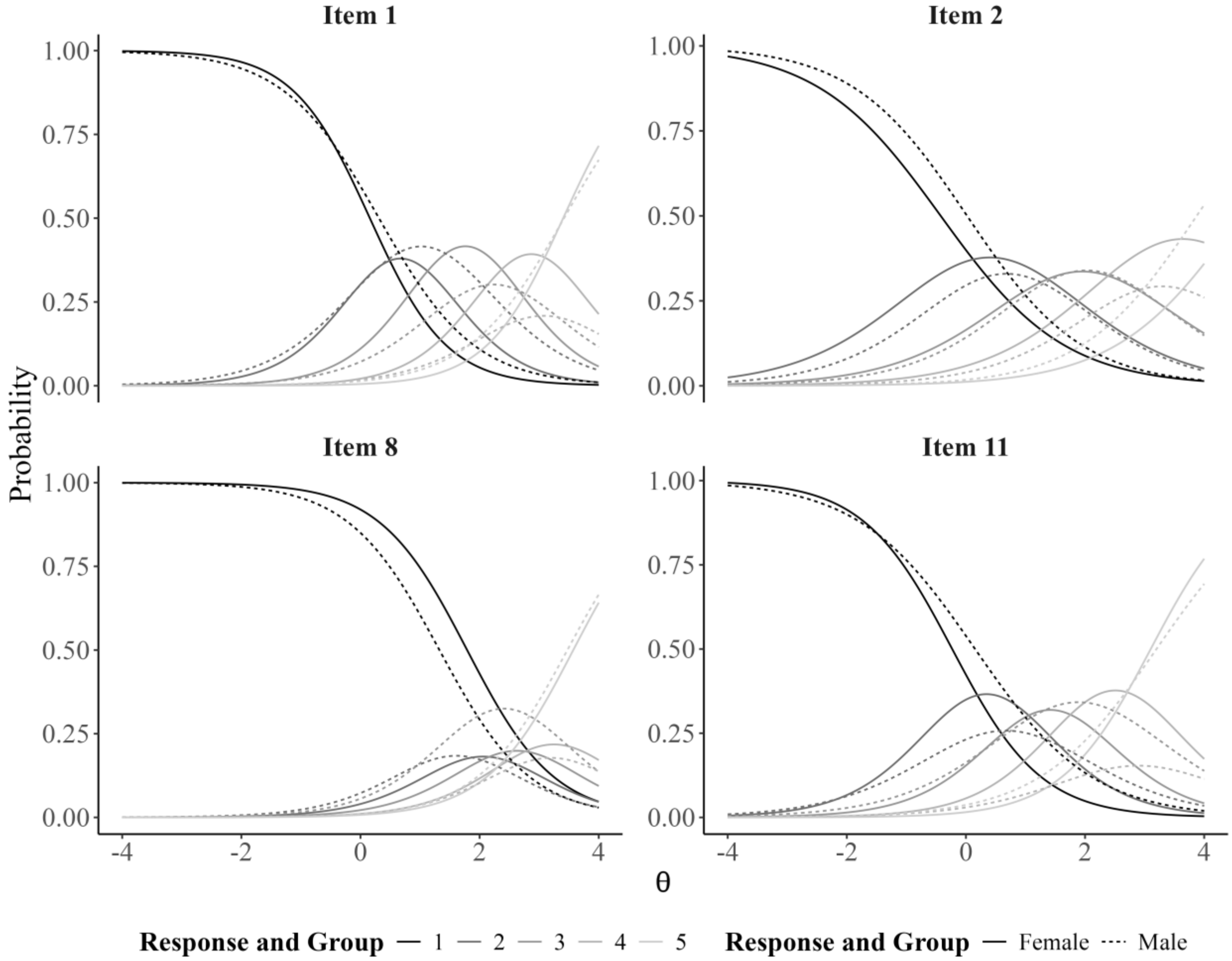
Supplemental Figure 2. Influential interactions that do not contain impact.



Note: Influential interactions for impact conditions that do not contain impact. Note: IRT = item response theory; RF Classification = random forest classification. Panel A illustrates the effect of the strength of DIF in the b parameter across sample types on sensitivity. As DIF strength increases, sensitivity improves for the focal group while declining for the reference group and full sample. Panel B displays the effect of DIF strength across classification methods on sensitivity. RF Classification models maintain relatively stable sensitivity, while IRT models exhibit decreased sensitivity as DIF strength increases. Panel C shows sensitivity by classification method across sample types, revealing that RF models produce more consistent sensitivity across samples, while IRT models show substantially lower sensitivity for the full and reference groups. Panel D presents the precision of classifications by sample across DIF strength levels. Precision remains highest for the reference group, while focal group precision decreases with increasing DIF strength, regardless of classification method.

Supplemental Figure 3.

Trace lines for items with gender-based Differential Item Functioning (DIF) on the Impairment subscale of the Duke Misophonia Questionnaire (DMQ; Rosenthal et al., 2021).



Note: This figure displays item characteristic trace lines for the items from the Impairment subscale of the DMQ that showed evidence of gender-based DIF. Trace lines represent the probability of endorsing each response category as a function of the latent trait (θ), plotted separately for male and female respondents. Differences between gender-specific curves suggest that individuals with equivalent levels of underlying impairment may respond differently to these items based on gender.